CROSSSAFE · CONCEPT OPTIONS

CrossSafe concept options

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DOCUMENT CONTROL

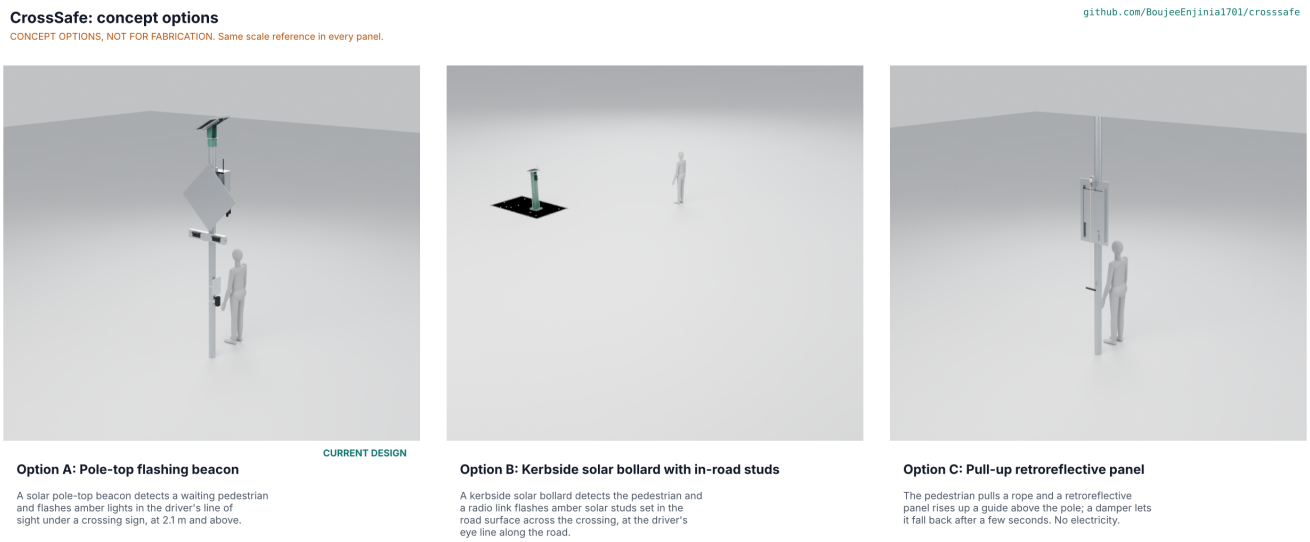
Document ID	CRS-OPT-001	Version	0.1
Status	Draft	Date	2026-10-31
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/crosssafe		
Source	docs/02a-concept-options.md @ d9f28be		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	First concept options (kit 1.9.1 upgrade): the TRL 2 pole-top beacon as option A and two options with different working principles; each modelled at D3 in cad/src/model.py; scored against CRS-URS-001; proposed, awaiting Amish

CrossSafe concept options

Option A (the pole-top flashing beacon, the design of the TRL 2 and TRL 3 work) scores 138 of 185 and option C (a pull-up retroreflective panel on a rope) 128, which is 7 % lower and so inside the 10 % band: both go to Amish without a pick, with A as the design that continues unchanged until he decides (Proposed, awaiting Amish; CRS-DDR-004). Option B (kerbside bollard with in-road studs) scores 108, 22 % lower, and is not recommended. Method: STANDARDS section 21. All three options are modelled to D3 in `cad/src/model.py` and pass `cad/fidelity.json`.



The options

Option A: Pole-top flashing beacon

Working principle: a pedestrian-activated, solar-powered rapid-flash lamp on the crossing sign. This is the TRL 2 design after the changes of CRS-DDR-001 to CRS-DDR-003 and CRS-DDR-005; it does not change here. A button and a radar woken by a PIR detect a waiting pedestrian, a LoRa link starts the matching beacon across the road, and two amber indications per face flash in the IA-21 wig-wag pattern for the crossing time. A 20 W panel and a 12.8 V 12 Ah LiFePO4 battery in a ventilated, shielded IP65 box power it (16 Wh/day, 7.7 days of autonomy). 117 modelled parts, 27.6 kg, $364.00 in parts against the $350 target. Key risks: cost over target (PR-015), flash area (PR-016), detection quality and visibility that only a trial can settle (PR-002, PR-005). Needs from a partner: the local rules, a site, and pole permission.

Option B: Kerbside solar bollard with in-road studs

Working principle: warn drivers at road level instead of at eye height. A steel bollard 1.25 m high at the kerb carries the button, the radar, the controller, the battery and a 10 W panel; eight amber solar road studs set across the crossing flash on command over the radio link, in a wave towards the pedestrian. Each stud (cast aluminium body, polycarbonate lens, 18650 cell, solar tile, four LEDs) is self-powered and a wheel passes over it. 156 modelled parts, 45.5 kg. Cost is about $400 per crossing side (estimate: $96 of controls reused from the BOM of A, $15 panel, $30 battery, about $58 for the steel bollard, galvanizing and anchors, and eight studs at about $25, which is a guess for a road-rated stud with radio; a consumer solar stud is about GBP 21 in a pack of four and is not rated for traffic).

Strengths: no pole needed, hard to steal the electronics, no overhead sign. Weaknesses: studs at road level are seen at a shallow angle and are not a form the pilot jurisdictions list, a road works permit is needed to cut the studs in, the energy per stud is thin (S19), and every stud is a wear item.

Option C: Pull-up retroreflective panel

Working principle: no electricity at all. The pedestrian pulls a rope; a pulley lifts a 400 x 500 mm retroreflective warning panel up two angle guides in front of the sign so it rises into the driver's view; an end-of-travel hydraulic damper (the fix of CRS-DDR-005) stops it when it is released, and it falls back on its own weight. Retroreflective sheeting returns headlight beams to the driver at night; in daylight the moving, bright panel is a visual cue. 40 modelled parts, 10.9 kg, about $70 in parts (estimate: $25 panel with sheeting, $18 backing plate and guides, $9 two saddles and bands, $8 pulley, pin, rope and handle, $6 damper, $4 clip and fixings). Strengths: cheapest, nothing to charge, almost nothing to steal. Weaknesses: it works only when a pedestrian pulls it, so it does nothing for the pedestrian who cannot reach or pull, does not warn the far side, gives no confirmation and no flash, and has no evidence of effect on yielding; it is a new device for a road authority to accept.

First-order numbers

Table 1. Options compared. Costs of B and C are estimates; A is the priced BOM.

Quantity	Option A	Option B	Option C
Estimated parts cost (USD)	364.00 (BOM, existing pole)	about 400 (estimate)	about 70 (estimate)
Mass (kg)	27.6	45.5	10.9
Output or throughput	4 amber indications, about 3,900 cd each (screening); 12.8 s per crossing	8 amber studs in the road, each 4 LEDs	one 400 x 500 mm retroreflective panel, raised on demand
Energy or effort	16 Wh/day, 7.7 days autonomy	about 15 Wh/day at the bollard; 0.8 Wh/day per stud	none; pull of about 20 N
Parts count (cad/fidelity.json)	117	156	40
STEM check	18 rows: 9 within, 8 plausible, 1 fails with a fix proposed (S15); S3 and S5 fixed	3 rows: 2 within, 1 plausible (S19 thin)	3 rows plausible; S22 fixed

Scoring

Table 2. Weighted scoring (weights: High UR 3, Medium 2; scores 1 to 5, 5 best). Maximum 185.

Criterion	Source	Weight	A	B	C	Reason
Drivers notice the warning and yield	UR-001	3	4	3	2	RRFB-type flashing has measured effect (FHWA); studs sit low; a panel has no flash
Warning when someone waits, both sides	UR-002	3	4	4	2	A and B detect and sync over radio; C needs a pull and does one side
Anyone can operate it and see it is on	UR-003	3	4	4	3	A and B have button, tone and light; C is a rope at 1.15 m with no confirmation
Warning lasts long enough and no longer	UR-004	3	4	4	2	A and B are timed by firmware; C falls when released
Works without mains in poor sun	UR-005	3	4	3	5	A has 7.7 days; B's studs are thin; C needs no power
Meets local rules and survives weather	UR-006	3	4	2	2	A follows IA-21 form; B and C are not forms the rules list
Fault never leaves a misleading warning	UR-007	3	4	3	4	A has a fault log and timers; C fails to dark; B has eight points of failure
Seen early enough to stop	UR-014	3	4	2	3	A seen from 100 m; studs at a shallow angle; C only at night by headlights
Costs little to buy	UR-009	3	2	1	5	$364 over $350; about $400; about $70
Fits an existing pole, installs quickly	UR-008	2	3	2	4	A 120 min; B needs a base and road works; C two saddles
Built and repaired with ordinary tools	UR-010	2	3	2	5	117, 156 and 40 modelled parts
No images or audio kept	UR-011	2	5	5	5	No camera in any option
Resists theft and vandalism	UR-012	2	4	2	3	A electronics at 2.8 m; studs and bollard at the kerb; C has a cheap pull handle
Lights not harmful to look at	UR-013	2	3	4	5	A flash area At risk (PR-016); C has no light
Weighted total			**138**	**108**	**128**	A 74.6 %, B 58.4 %, C 69.2 % of 185

The scoring is the author's judgment from the first-order numbers; the ranking of A and C depends on the weight of cost (UR-009) against evidence of effect (UR-001, UR-002, UR-004). With cost weighted 1, A leads C by 12 % (134 against 118); with cost weighted 4, by 5 % (140 against 133); A leads at every weight, but the gap is inside the 10 % band at weights of 2 and above, which is why both go to Amish.

Recommendation

Both A and C go to Amish without a pick, because they are within 10 %. A is recommended as the continuing design on the evidence of effect and function, and C is the cheap alternative for crossings that cannot fund A. The runner-up's best idea to graft onto A is C's cheap pull-up form as a low-cost "tier 0" for a crossing with no budget; and the damper of C's end stop is a useful part for the light bar's lid. B is not recommended. The choice is recorded as "Proposed, awaiting Amish" in CRS-DEC-001 and argued in CRS-DDR-004.